

Research Statement

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Air pollution, road deaths, and groundwater depletion share one feature. Whoever causes the damage does not pay for it. Environmental and resource economics calls this an externality. Governments respond with zoning, subsidies, traffic rules, and infrastructure. Whether these policies work, and what else they change, is an important empirical question. My research evaluates them with causal inference and large-scale administrative data, and the results often differ from what the policy assumed. Externalities arise not only from factory chimneys but also on farmland and city streets, so my work falls into two lines, one on farms and one on cities.

1. Land, Food, and the Environment

Agriculture is the largest user of land and water, a major source of emissions and nutrient runoff, and the livelihood of millions of households. A policy that cleans up farming at the farmer's expense will not be adopted. Must environmental quality and farm income trade off? Often not.

Solar power is treated as a rival for farmland. [1] and [2] show that panels on existing farm buildings displace nothing. Chicken farms that adopt them sell 5.8% more, and the gain comes from higher output rather than higher quality. The conflict is about where panels go, not whether they go up.

Siting can be set by policy. Taiwan zones land for aquavoltaics, which places solar panels above working fish ponds instead of converting farmland. [3] finds that farmland prices rose 13.5% after land entered these zones. The increase likely reflects the value of the development rights the zoning creates.

Fertilizer and pesticide runoff resists regulation because it comes from tens of thousands of sources no inspector can reach. [4] finds that farms adopting food traceability are 8.2, 15.4, and 40 percentage points less likely to use chemical fertilizer, pesticide, and groundwater. Revenue rises too. A food safety system did what enforcement cannot.

Biodiversity disappears from private land because nobody pays for it. [5] finds that forest farms with greater tree diversity earn less on average but face far steadier revenue, and the reduction in

risk outweighs the loss in income. Diversity pays in stability rather than in income.

2. Cities, Movement, and Externalities

Transport emissions keep rising, and transport kills directly. Cities answer with rail, fare subsidies, and traffic rules. None is cheap, and the effect of none is known in advance.

[6] studies the removal of the two-stage left turn rule for motorcycles in part of Tainan City. Accidents fell 21%, victims 19%, and vehicles involved 28%. Road safety usually means building something. Here a rule change, at a cost that is trivial next to any construction project, worked just as well.

Building does pay when it is aimed at the right people. [7] evaluates Taiwan's nationwide upgrade of pedestrian-protective infrastructure at intersections, such as refuge islands and relocated crosswalks. Injury crashes fell 8.2% in treated townships and pedestrian crashes 14.8%. The gains are concentrated at narrow single-lane intersections. Within eight months, each NT\$1 spent returned about NT\$3.5 in avoided medical costs.

Metro ridership responds quickly to shocks from outside the system, whether a policy or a pandemic. On the policy side, [8] finds that the Taipei monthly pass raised ridership, passenger kilometers, and trip value by 4.3%, 4.6%, and 4.6%. Fare subsidies are now sold as climate policy, but emissions fall only if 70% to 80% of the new trips are diverted from cars and motorcycles, so ridership alone does not show whether the subsidy worked. On the pandemic side, [9] finds that each new COVID-19 case reduced Taipei Metro use by 1.43%, even though Taiwan never imposed a lockdown. Perceived health risk alone was enough to move riders off the system.

Mass events send the bill to people who never attended. [10] finds that accidents rose 14.9% and victims 17.4% during the Dajia Mazu Pilgrimage, and that accidents rose 9.8% even in townships off the route. [11] measures the air quality cost of Hakka tomb sweeping at the Lantern Festival. Neither cost enters the planning of these events.

3. Other

[12] builds a strategic trade model with green investment. The Carbon Border Adjustment Mechanism pushes exporters toward cleaner technology, though its market structure effects vary by industry. [13] finds that consumers shifted food purchases online under pandemic risk. [14] finds that hotel revenue recovered through higher prices rather than more guests, and that restoring the labor supply would raise revenue by up to 17.8%.

Farms or cities, the question is the same. When a policy changes the price of harming the environment, or the reward for improving it, who changes their behavior, by how much, who bears the cost, and who receives the benefit.

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